# Aeronautics and Space Technology Club

## IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 09 · Part C: Electronics for Signal Processing Level 3

**Track:** Circuit Noise, Bridge Sensors, Amplifier Design, Lock-In Detection, Noise Budgets

**Prerequisites:** Problems 07, 08 · **Estimated effort:** 12–16 hours

**Sensor Conditioning and Circuit Noise: Getting a Microvolt Signal Out of a Noisy World**

A strain gauge delivers millivolts, a photodiode picoamperes, and both sit next to a 50 Hz mains field a million times larger. Compute the fundamental noise floor of resistors and diodes, size a bridge amplifier and a transimpedance amplifier, build a digital lock-in that recovers a tone buried 40 dB under noise, and assemble an end-to-end noise budget that tells you where the next decibel is cheapest.

## General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

| Stage | Phase | What you should have at the end of it |
|---|---|---|
| 1 | Think (∼20%) | Theory understood; every analytic result derived; signals, units and sampling rates written down. |
| 2 | Build (∼30%) | Core code working end to end on the baseline data and validated against the analytic results. |
| 3 | Experiment (∼30%) | Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers. |
| 4 | Refine (∼20%) | Weaknesses fixed, honest analysis and memo written, repository folder clean. |

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/09_sensor_conditioning_and_circuit_noise/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random

seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of $2\pi$, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

## Background and Motivation

The best digital algorithm cannot recover information that the analog front end never delivered. The front end must amplify a tiny sensor signal enough to use the ADC range, without adding more noise than the sensor itself, and without letting interference through. Three facts govern the design: every resistor produces thermal noise, the first amplifier stage dominates the cascade's noise, and a signal at a known frequency can be extracted from enormous broadband noise by narrow-band detection (a lock-in amplifier), at the cost of measurement time.

You will quantify each of these with numbers, then combine them into a noise budget for a strain-gauge channel on a wing spar and a photodiode channel of a sun sensor. The digital lock-in you write is the same algorithm used in laboratory instruments, in modulated optical sensors and in spread-spectrum receivers.

## Theoretical Framework

### 1. Fundamental Noise

Thermal (Johnson) noise of a resistance $R$ at temperature $T$ has open-circuit voltage density $e_n = \sqrt{4k_\mathrm{B}TR}$ ($4.0\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ for $1\,\mathrm{k\Omega}$ at $290\,\mathrm{K}$); shot noise of a current $I$ has density $i_n = \sqrt{2qI}$ ($0.57\,\mathrm{pA}/\sqrt{\mathrm{Hz}}$ at $1\,\mu\mathrm{A}$). Densities of independent sources add in power; RMS noise over a bandwidth is $v_\mathrm{rms} = e_n\sqrt{B_n}$ with $B_n$ the *noise bandwidth*, which for a first-order low-pass with corner $f_c$ is $(\pi/2)f_c$ and for a first-order lock-in output filter with time constant $\tau$ is $1/(4\tau)$.

### 2. Amplifier Noise

An amplifier is modelled as a noiseless one with an input noise voltage $e_{na}$ and input noise current $i_{na}$. With source resistance $R_s$ the total input-referred density is

$$e_\mathrm{tot}^2 = 4k_\mathrm{B}TR_s + e_{na}^2 + i_{na}^2R_s^2, \qquad R_{s,\mathrm{opt}} = \frac{e_{na}}{i_{na}}. \tag{1}$$

For a cascade the input-referred noise of stage $k$ is divided by the squared gain of all earlier stages, so the first stage dominates.

### 3. Sensors and Their Front Ends

A quarter Wheatstone bridge with one active gauge of nominal resistance $R$ and strain $\varepsilon$ ($\Delta R/R = \mathrm{GF}\,\varepsilon$) and excitation $V_{\mathrm{ex}}$ gives

$$V_{\mathrm{out}} = V_{\mathrm{ex}}\left[\frac{R+\Delta R}{2R+\Delta R} - \frac{1}{2}\right] \approx \frac{V_{\mathrm{ex}}}{4}\,\frac{\Delta R}{R}. \tag{2}$$

It is amplified by an instrumentation amplifier of gain $G = 1{+}2R_f/R_g$, whose common-mode rejection ratio keeps mains pickup out. A photodiode is read by a transimpedance amplifier $V = -I_{\mathrm{ph}}R_f$, which adds the current noise of the feedback resistor $i_n = \sqrt{4k_{\mathrm{B}}T/R_f}$.

### 4. Lock-In Detection

To measure the amplitude $A$ and phase $\phi$ of $x(t) = A\cos(\omega_0 t + \phi) + n(t)$ multiply by $2\cos\omega_0 t$ and $2\sin\omega_0 t$ and low-pass filter:

$$X \approx A\cos\phi, \qquad Y \approx -A\sin\phi, \qquad R = \sqrt{X^2+Y^2}, \quad \theta = \mathrm{atan2}(-Y, X). \tag{3}$$

For white noise of per-sample standard deviation $\sigma$ at sample rate $f_s$ and a first-order output filter of time constant $\tau$, each quadrature has noise standard deviation $\sigma_X \approx \sigma/\sqrt{f_s\tau}$: the amplitude resolution improves as $1/\sqrt{\tau}$ until $1/f$ drift dominates.

## Data and Simulation Specification

| | |
|---|---|
| **Resistor noise** | $R = 350\,\Omega, 1\,\mathrm{k}\Omega, 100\,\mathrm{k}\Omega, 1\,\mathrm{M}\Omega$ at 290 K, bandwidth 10 kHz. |
| **Strain bridge** | Quarter bridge, $R = 350\,\Omega$, gauge factor 2.0, $V_{\mathrm{ex}} = 5$ V, strain 0–2000 $\mu\varepsilon$, ADC range $\pm 2.5$ V. |
| **Photodiode** | Photocurrent 1 nA–100 µA, feedback $R_f = 10\,\mathrm{M}\Omega$, diode capacitance 20 pF, feedback capacitor 1 pF. |
| **Lock-in data** | `sp_common.synth.lockin_signal(fs=10000, duration=20, f_ref=137, amp=1e-3)`: 1 mV tone, white noise, $1/f$ noise and 50 Hz pickup of amplitude 0.1 V; input SNR about −40 dB. |
| **Real data (optional hardware)** | Photodiode and LED chopped at a known frequency by a microcontroller, sampled with the microcontroller ADC; or a loudspeaker tone recorded next to a running fan with the sound card and recovered by your lock-in. |

## Your Task

1. **Noise floors (Stage 2).** Compute $e_n$ for the four resistors and the RMS noise in 10 kHz; compute the shot-noise density for currents from 1 nA to 1 mA. Verify numerically by generating white noise with the computed density and measuring its RMS in a band. Plot the thermal noise of resistors against $R$ and mark the crossing with a typical op-amp $e_{na} = 10\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$.
2. **Strain-gauge channel.** Evaluate Eq. (2) exactly and in the small-signal limit for 500 $\mu\varepsilon$. Choose the instrumentation-amplifier gain so that 2000 $\mu\varepsilon$ fills the ADC range, then compute the output noise of gauge, amplifier ($e_{na} = 10\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$) and a 16-bit ADC in a 100 Hz bandwidth and the resulting strain resolution in micro-strain.
3. **Cascade noise and optimum source resistance.** Model a two-stage amplifier with gains $G_1, G_2$ and noise densities $e_1, e_2$; show that $e_{\mathrm{tot}}$ is dominated by stage 1 for $G_1 \gg 1$. Plot the noise figure against $R_s$ for $e_{na}, i_{na}$ values of your choice and verify $R_{s,\mathrm{opt}} = e_{na}/i_{na}$.

4. **Transimpedance amplifier.** For $R_f = 10\,\mathrm{M\Omega}$ compute the current-noise density, the signal bandwidth set by $R_f C_f$ and the SNR of a 1 nA photocurrent in that bandwidth. Plot the SNR versus $R_f$ and choose $R_f$ for a 10 µA full-scale current with a 2.5 V output limit.
5. **Digital lock-in.** Implement Eq. (3) with first- and second-order Butterworth output filters. Recover amplitude and phase of the synthetic tone for time constants $\tau = 0.25$ to 4 s and plot the RMS amplitude error against $\tau$ on log–log axes together with the $1/\sqrt{\tau}$ law; determine where $1/f$ noise or the 50 Hz leakage takes over.
6. **Alternatives.** Recover the same tone with (a) the FFT peak of the full record and (b) a narrow IIR band-pass. Compare accuracy, latency and the effect of choosing $f_{\mathrm{ref}}$ as a multiple of 50 Hz versus 137 Hz.
7. **Noise budget.** Assemble the strain-gauge noise budget (gauge, amplifier, aliasing of out-of-band noise, ADC quantisation) as a table in $\mu\varepsilon_{\mathrm{rms}}$ and identify the dominant term. State which single change (gain, bandwidth, excitation voltage, ADC bits) buys the largest improvement per unit cost.

**Self-check targets**

$e_n = 4.0\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ for 1 kΩ ($2.4\,\mathrm{nV}/\sqrt{\mathrm{Hz}}$ for 350 Ω); the quarter bridge gives 1.249 mV exactly and 1.25 mV in the small-signal limit at 500 $\mu\varepsilon$, so a gain of $G \approx 500$ fills a ±2.5 V range at 2000 $\mu\varepsilon$ (5 mV). A 10 MΩ feedback resistor has $i_n = 40\,\mathrm{fA}/\sqrt{\mathrm{Hz}}$. The lock-in recovers 1 mV to about 10% with $\tau \approx 1$ s and its amplitude error falls as $1/\sqrt{\tau}$ before drift dominates.

## Aerospace Application Link

Strain gauges on a wing spar measure flutter in a ground vibration test, and a sun sensor reads the position of the Sun on a quadrant photodiode in the presence of Earth albedo. For the strain channel report the minimum detectable strain (SNR = 1) in a 100 Hz bandwidth. For the sun sensor, explain how chopping the source (or modulating an on-board LED for a calibration reference) and using a lock-in rejects the slowly varying albedo background, and compute the integration time needed to resolve 0.1% amplitude changes if the white noise is $10^{-3}$ of the signal per $\sqrt{\mathrm{Hz}}$.

## Deliverables

**D1** **Noise floors:** thermal and shot noise tables, the Monte Carlo verification and the resistor-noise plot.

**D2** **Strain channel:** bridge equation check, gain selection, strain resolution calculation.

**D3** **Cascade and TIA:** first-stage dominance demonstration, noise figure versus $R_s$, TIA design with SNR versus $R_f$.

**D4** **Lock-in:** notebook with amplitude and phase recovery, the error-versus-$\tau$ plot and the comparison with the FFT and IIR alternatives.

**D5** **Noise budget:** the table with the dominant term and the recommended improvement.

**D6** **Memo (250 words):** "Where should the next decibel of strain-gauge SNR come from?" Cite at least two external sources.

**Bonus** (a) Add an AC-coupled input and a $1/f$ corner to the amplifier noise model. (b) Implement a dual-phase lock-in with a reference recovered from the signal itself (PLL). (c) Build the chopped-photodiode experiment and compare the measured noise density with your prediction.

# Learning Resources

### Circuit Noise and Low-Noise Design

- Horowitz, P., Hill, W. *The Art of Electronics*, chapter on low-noise techniques (noise sources, op-amp noise, noise figure).
- Motchenbacher, C.D., Connelly, J.A. *Low-Noise Electronic System Design.*
- Analog Devices, *Op Amp Noise* tutorial MT-047 (free).

### Sensors and Lock-In Detection

- Hoffmann, K. *An Introduction to Measurements Using Strain Gages*, HBM (bridge circuits).
- Stanford Research Systems, *About Lock-In Amplifiers* application note (free).
- Meade, M.L. *Lock-in Amplifiers: Principles and Applications.*

### Tools

- SciPy `scipy.signal` (`butter`, `sosfilt`): https://docs.scipy.org/doc/scipy/reference/signal.html

# Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

| Criterion | What to look for | Marks |
|---|---|---|
| **Conceptual depth** | Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour. | /20 |
| **Implementation quality** | Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass. | /20 |
| **Aerospace-application integration** | The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires. | /20 |
| **Analysis and critical thinking** | Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases. | /20 |
| **Communication** | Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow. | /20 |

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.